

Advanced Econometrics

Video for Lab 09

Notes for video

A random walk process

$$\underline{y_t} = \underline{1} y_{t-1} + \epsilon_t$$

$$\alpha = 0 \quad y_t = \alpha \cdot y_{t-1} + \epsilon_t$$

$\alpha \in (-1, +1)$ $y_t = 0 \cdot y_{t-1} + \epsilon_t$ $y_t = \epsilon_t$ $AR(1)$ is stationary

white-noise process

Autoregressive processes

$$AR(1) \quad y_t = \alpha \cdot y_{t-1} + \epsilon_t$$

$$\alpha \in (-1, +1)$$

$AR(1)$ is stationary

AR(2) ~~$\alpha \in (-1, +1)$~~
general version . does not apply here!

$$y_t = \alpha y_{t-1} + \beta y_{t-2} + \epsilon_t$$

$$y_t = \beta y_{t-2} + \epsilon_t$$

$$AR(p) \quad y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \dots + \alpha_p y_{t-p} + \epsilon_t$$

Random walk with a drift

RW with an intercept (or constant)

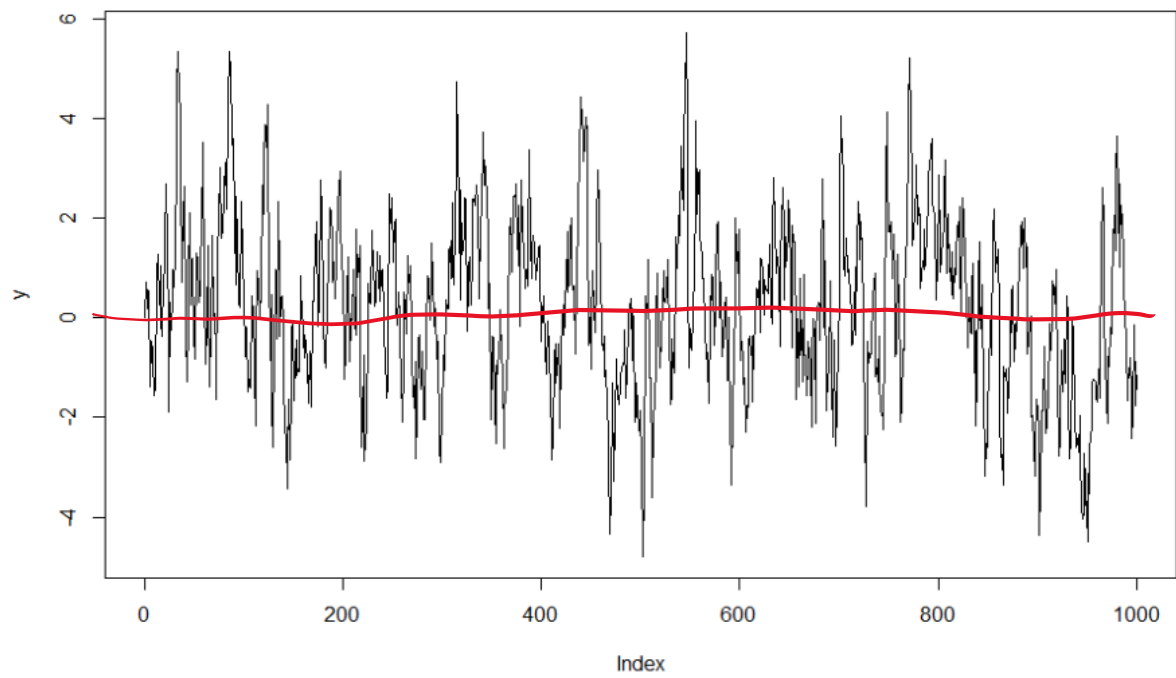
AR with a constant

$$AR(1) \quad y_t = \alpha_0 + \alpha_1 y_{t-1} + \epsilon_t$$

$$E[y_t] = \frac{\alpha_0}{1 - \alpha_1}$$

RW with a drift

$$y_t = y_{t-1} + \mu + \epsilon_t$$



type III DR

RW with a drift and a trend $\rightarrow$ accumulates
to quadratic trend

$\alpha \in (-1, 1)$

trend stationary variable

if we removed the linear trend
the remaining part would be
stationary

the
whole process
is
non stationary

